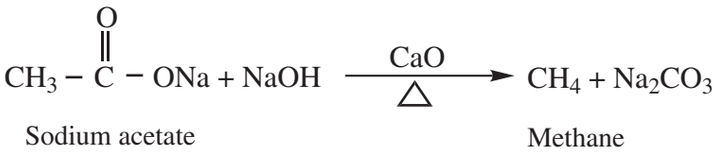
2) Decarboxylation

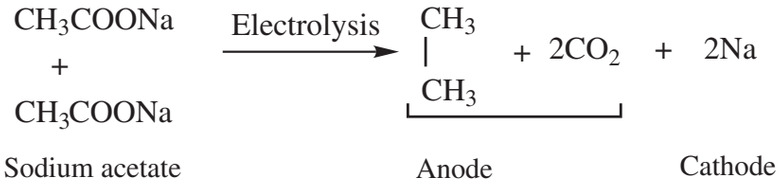
Removal of CO_2 from carboxyl group is called as **decarboxylation.** Carboxylic acids lose carbon di oxide to form hydrocarbon when their sodium salts are heated with soda lime (NaOH and CaO in the ratio 3: 1)

Example



3) Kolbe's electrolytic decarboxylation

The aqueous solutions of sodium or potassium salts of carboxylic acid on electrolysis gives alkanes at anode. This reaction is called kolbes electrolysis.

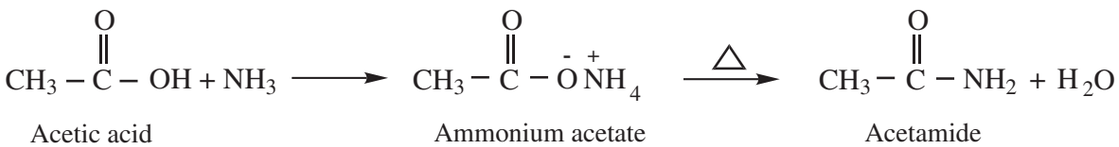


Sodium formate solution on electrolysis gives hydrogen

4) Reactions with ammonia

Carboxylic acids react with ammonia to form ammonium salt which on further heating at high temperature gives amides.

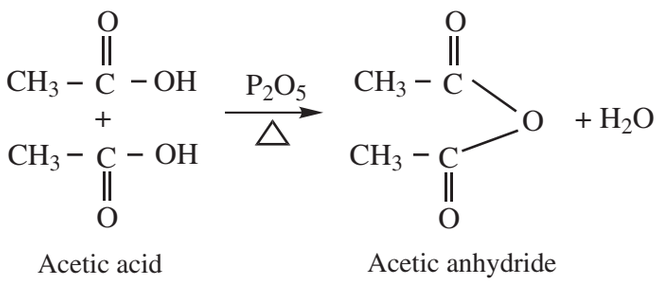
Example



5) Action of heat in the presence of P_2O_5

Carboxylic acid on heating in the presence of a strong dehydrating agent such as P_2O_5 forms acid anhydride.

Example



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